

Arunabha Majumder

Robotics Engineer — Mechanical Design, Motion Control, Actuators & Sensors, Hands-On Builds

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SUMMARY

Robotics engineer who builds machines that work. Designed, built and tested robotic systems from concept prototype to working hardware: mechanical design in SolidWorks, 3D-printed iterations, integration of motors, actuators and sensors, motion control, and test rigs. Leads on mechanical and mechatronics, and happy to pitch in on electronics and software.

THINGS I'VE BUILT

- **Shoulder exoskeleton (PhD):** Hybrid-actuated robotic joint (parallel spring + 6 Nm motor) with sensors, embedded control and test rigs; cut energy use by more than 25% (Best Paper Award, 2026).
- **5-bar parallel robot setup (PhD):** Kinematic model and sensorless force estimation, validated to below 10% RMS error.
- **Pneumatic actuator bench (MTech):** Pneumatic artificial muscles and a full test rig (valves, pressure sensors, DAQ), built from raw materials.
- **Rehabilitation exoskeleton (Bachelor's):** Designed, machined, assembled and user-tested; Best Major Project Award.
- **Go-Kart (2017):** Designed, fabricated and assembled from the ground up.

EXPERIENCE

Research Engineer – Robotic Machines & Motion Control (PhD Project: VIEXO) | 2023 – Present *Aalborg University, Aalborg, Denmark*

- Took a robotic machine from concept prototype to a working system tested with people by leading the mechanical design in SolidWorks and iterating through 3D-printed prototypes; the design cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Integrated motors, actuators and sensors (6 Nm motor, encoder, force sensors, DAQ and embedded control) into one machine, working across mechanical, electronics and control.
- Reduced user muscle effort by more than 15% by implementing motion control (Assist-as-Needed controller) on the hardware, then testing and refining it with 12 participants.
- Achieved below 10% RMS force-estimation error on a 5-bar planar parallel robot, validated against a force sensor, by modelling its kinematics (Jacobian) and variable-stiffness actuators.
- Built custom test rigs and tracked down mechanical, electrical and control issues on the bench; mentored Bachelor's (B.Tech) students (2023 – 2025).

MTech Researcher – Pneumatic Actuator Control (Master's Thesis) | 2020 – 2022 *CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India*

- Reduced tracking error to 0.3–0.78% by building pneumatic actuators and their test rig from scratch (valves, pressure sensors, DAQ) and developing a neural-network-based controller.

Bachelor's Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019 *Siddaganga Institute of Technology, Tumakuru, India*

- Won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant by leading CAD design, machining, hands-on assembly and user testing.

SKILLS

- **Mechanical & CAD:** SolidWorks (CSWA), mechanical design, assemblies, drawings
- **Motion control & sensors:** Motors, encoders, actuators, force and pressure sensors, DAQ, PID and model-based control, kinematics

- **Fabrication:** 3D printing (Bambu Lab, Ultimaker Cura); machining and assembly (Bachelor's)
- **Software & embedded:** Python, MATLAB/Simulink, Arduino, LabVIEW (basic)
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA); two peer-reviewed publications (ISRM 2026, IEEE CONECCT 2022)

REFERENCES

Available upon request.